HNS-36 Internal PoC Report

Prepared for External Validation

Coordinate-Based Structural Diagnosis of AI-Generated Reasoning

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Primary Claim	HNS-36 can function as a diagnostic language for structural reasoning errors in AI outputs.

Core conclusion

HNS is not a text generation tool. It is a diagnostic language that assigns names and coordinates to structural problems in AI-generated reasoning.

1. Executive Summary

This report presents a preliminary proof-of-concept for HNS-36 as a coordinate-based diagnostic layer for AI-generated reasoning. The experiment compares ordinary content-level critique with HNS-36 structural diagnosis across five representative AI-output cases: digital loneliness, educational inequality, climate inaction, organizational resistance to change, and social media addiction.

The central finding is that HNS-36 does not merely judge whether an answer is vague, incomplete, or overgeneralized. It assigns a named structural error type and an HNS coordinate to the location of the reasoning failure. This allows AI-output problems to be recorded, compared, and accumulated in a systematic format.

Finding	Meaning for HNS
HNS produced named structural diagnoses in all five cases.	The framework operates as a diagnostic language, not only as an interpretive philosophy.
Unsupported Causality appeared in all five cases.	AI outputs frequently assert causal transitions without providing bridge mechanisms.
Metaphor Contamination was detected in cases involving biological or neurochemical explanations.	HNS can name layer-mixing errors that generic critique usually labels only as oversimplification.
Layer Jump and Scope Drift appeared repeatedly.	AI explanations often move between individual, relational, and societal levels without transition logic.
The report defines HNS as a backend verification layer.	HNS is best positioned as an internal structural diagnostic system, while natural language remains the user-facing layer.

Scope of evidence

The present report is preliminary. It is based on five cases and a single evaluator. It demonstrates a plausible diagnostic capability and a repeatable record format, but it does not yet establish statistical validity or external reliability.

2. Problem Statement

Modern AI systems often produce fluent explanations that appear coherent at the surface level but contain hidden structural defects. These defects include missing causal mechanisms, unmarked jumps between human layers, shifts from individual-scale claims to society-scale conclusions, and the use of biological or physical metaphors to explain psychological or intentional behavior.

Generic critique can identify such problems in broad language, for example by saying that an answer is vague, unsupported, overgeneralized, or circular. However, generic critique usually lacks a coordinate system. It does not consistently identify where the error occurs within a layered human-structure model, which categories have been conflated, or which transition logic is missing.

PoC question

Can HNS-36 provide a more specific, reproducible, coordinate-based diagnosis of AI-output reasoning errors than generic critique?

3. HNS-36 Diagnostic Framework

For this PoC, HNS-36 is treated as a structural coordinate system for diagnosing reasoning defects in AI-generated text. Each detected issue is assigned both an error type and a coordinate or transition path. The purpose is not to rewrite the output, but to identify the structural location and nature of the reasoning failure.

Error Type	Operational Definition	Typical Signal
Layer Jump	Movement across HNS layers without explicit bridge logic.	A sentence moves from perception to intention, or from society to internal psychology, without explaining the transition.
Scope Drift	Shift between individual, relational, organizational, or societal scale without transition logic.	The answer begins with personal experience and ends with a broad societal claim.
Unsupported Causality	Assertion of a causal relationship without a mechanism.	The answer says “A causes B” but does not explain how A produces B.
Metaphor Contamination	Use of L1 Physical or biological facts to explain L3/L4 psychological or intentional phenomena without justification.	The answer uses dopamine, brain development, or evolution as a direct explanation for behavior.
Category Ambiguity	Conflation of distinct HNS cognitive categories, such as interpretation, intention, action, and evaluation.	A concept such as “opportunity” or “investment” mixes motivation, access, institutions, and normative judgment.

4. Experimental Methodology

Item	Description
Objective	Verify whether HNS-36 can diagnose structural problems in AI-generated text more specifically and systematically than generic critique.
Test cases	Five representative explanatory prompts: digital loneliness, educational inequality, climate inaction, organizational resistance to change, and social media addiction.
AI response generation	Typical AI responses generated without HNS awareness.
Evaluation axis 1	Generic critique without HNS: content-level problem identification.
Evaluation axis 2	HNS-36 diagnosis: error type, coordinate, and diagnostic explanation.
Aggregation	Frequency count by error type and HNS-specific detection count.
Primary limitation	Single evaluator and five cases only; no external rater in this edition.

4.1 Hypotheses

Hypothesis	Statement	Expected Indicator
A	HNS diagnosis provides more specific error names and coordinates than generic critique.	Each issue can be expressed as a named error type plus an HNS coordinate or transition.

Hypothesis	Statement	Expected Indicator
B	HNS-specific error types are difficult or impossible to name by generic critique alone.	Metaphor Contamination and layer-transition errors appear as distinct HNS diagnoses.
C	Recurring structural error patterns exist across AI-generated outputs.	The same HNS error types recur across multiple domains.

5. Case-Level Results

The following table summarizes the five case diagnoses. The full internal report contains detailed source responses and line-level diagnostic notes. This prepared-for-external-validation edition consolidates the key structural patterns for easier review.

Case	Prompt Domain	Main HNS Errors	HNS-Specific Detections
1	Digital loneliness	Layer Jump; Unsupported Causality; Scope Drift	3
2	Educational inequality	Unsupported Causality; Category Ambiguity	3
3	Climate inaction	Layer Jump; Metaphor Contamination; Unsupported Causality	4
4	Organizational resistance to change	Scope Drift; Unsupported Causality; Layer Jump	4
5	Social media addiction	Category Ambiguity; Metaphor Contamination; Unsupported Causality; Scope Drift	4

5.1 Representative Diagnostic Examples

Source Claim	Generic Critique	HNS-36 Diagnosis
“Social media creates only shallow connections.”	Surface-level explanation; lacks detail.	Layer Jump: L5 Relational / C6 Interaction is used to explain L3 Internal loneliness without bridge logic.
“Humans are naturally short-sighted.”	Overgeneralization.	Metaphor Contamination: L1 Physical/evolutionary framing is used to explain L4 Intentional behavior without justification.
“The company culture is too rigid.”	Circular reasoning.	Unsupported Causality: L6 Societal / C1 Existence explains rigidity through rigidity without mechanism.
“Likes release dopamine and bring people back.”	Oversimplified neuroscience.	Metaphor Contamination: L1 Physical neurochemical fact is used to explain L4 intentional/habitual return behavior.

6. Aggregated Results

Error Type	Count	Cases	Interpretation
Unsupported Causality	9	All 5 cases	Most frequent error. AI explanations repeatedly assert causal transitions

Error Type	Count	Cases	Interpretation
			without mechanisms.
Layer Jump	4	Cases 1, 3, 4, 5	Frequent unbridged movement between perceptual, internal, relational, and societal layers.
Scope Drift	3	Cases 1, 4, 5	Repeated movement from individual explanation to societal conclusion without transition.
Category Ambiguity	4	Cases 2, 5	Ambiguous mixing of cognitive categories such as intention, action, evaluation, and existence.
Metaphor Contamination	2	Cases 3, 5	Biological or neurochemical terms used as direct explanations of psychological or intentional behavior.
Total	22	Average 4.4 per case	The small sample still shows recurring structural patterns.

Dimension	Generic Critique	HNS-36 Diagnosis
Problem naming	Vague, overgeneralized, correlation-causation confusion.	Named structural types such as Layer Jump, Scope Drift, and Metaphor Contamination.
Coordinate assignment	None.	Every issue can be assigned a coordinate or transition path.
Level of critique	Content-level: what seems wrong.	Structural-level: which layer/category relationship is defective.
Reproducibility	Evaluator-dependent.	Coordinate-based record supports accumulation and comparison.
System role	Editorial feedback.	Backend verification and diagnostic layer.

7. Interpretation

7.1 Why Unsupported Causality Matters

Unsupported Causality was the most frequent error type, appearing in all five cases. This suggests that fluent AI explanations often contain a missing middle: the answer states a plausible causal relationship but does not articulate the mechanism that connects the starting condition to the conclusion. HNS makes this missing middle visible by requiring transition logic between layers and categories.

7.2 Why Metaphor Contamination Matters

Metaphor Contamination is especially important because it is difficult to name using ordinary critique. A generic evaluator may say that “dopamine” is an oversimplified explanation, but HNS can specify the structural error: a physical or biological layer is being used to explain an intentional or internal phenomenon without a justified bridge. This makes the error recordable and comparable across cases.

7.3 Why Individual-to-Societal Drift Matters

Several AI responses moved from individual-level psychology to societal-level conclusion without intermediate relational, institutional, or environmental logic. This is a recurring source of persuasive but structurally incomplete explanations. HNS can identify this as Scope Drift or Layer Jump rather than merely calling the answer broad or unsupported.

8. Correct Role Definition of HNS

Recommended definition

HNS-36 should be presented as a coordinate-based diagnostic layer for detecting structural reasoning errors in AI-generated outputs. It should not be presented as a replacement for natural language generation or as a frontend user interface.

HNS Role	Description
Diagnoses structural problems	Records Layer Jump, Scope Drift, Unsupported Causality, Metaphor Contamination, and Category Ambiguity with names and coordinates.
Assigns unique error names	Provides a classification unavailable in ordinary content critique.
Operates as a backend layer	Functions as an internal verification layer that can support external oversight.
Does not output final user text	A separate natural-language layer should handle readable user-facing responses.
Does not replace domain evidence	HNS diagnoses structure; it does not by itself prove empirical facts.

9. Relevance to AI Safety and External Verification

This PoC is directly relevant to AI safety because many safety failures are not only factual errors. They are structural reasoning errors: unjustified transitions, misplaced explanations, conflated categories, and unmarked shifts in scope. HNS-36 can serve as a diagnostic module inside a broader external verification architecture.

Use Case	How HNS-36 Contributes
AI-output audit	Flags structural defects in generated reasoning before or after publication.
Model comparison	Compares frequency of structural errors across models, prompts, or domains.
Red-team analysis	Identifies recurring failure modes beyond factual hallucination.
External verification	Provides coordinate-based diagnostic records that can support independent review.
Standardization research	Defines a repeatable vocabulary for structural reasoning defects.

10. Limitations

Limitation	Implication	Required Next Step
n=5 cases only	The result is indicative, not statistically conclusive.	Expand to 30+ cases across multiple domains.
Single evaluator	Evaluator bias and prompt dependency are possible.	Use multiple blinded evaluators and compare agreement.
Manual or AI-assisted coordinate mapping	Diagnosis may still depend on evaluator judgment.	Develop an automated coordinate mapper and calibration guide.
HNS-36 theoretical basis not fully tested here	The experiment uses the 6x6 structure but does not prove its necessity.	Add a formal justification document for the 36-cell decomposition.
No independent human trust/readability assessment in this report	The report diagnoses structure, not user-facing readability.	Run separate frontend evaluation for revised outputs.

11. Development Roadmap

Phase	Deliverable	Purpose
Phase 1	Internal PoC report	Demonstrate the diagnostic idea and identify recurring error types.
Phase 2	30-case validation dataset	Generate stronger frequency evidence across domains.
Phase 3	Evaluator prompt and scoring rubric	Make HNS diagnosis more reproducible across models and human evaluators.
Phase 4	Blind external review	Measure whether third-party evaluators agree with HNS diagnoses.
Phase 5	Automated HNS coordinate mapper	Reduce manual dependency and move toward operational implementation.
Phase 6	EVA/HNS integration prototype	Use HNS as a diagnostic engine within an external verification architecture.

12. Proposed Dataset Schema for Next Validation

The next PoC should be stored as both a human-readable report and a machine-readable dataset. The following schema is recommended for CSV or spreadsheet use.

Column	Description
Case ID	Unique identifier for each test case.
Domain	Topic domain such as education, climate, AI safety, organization, or health communication.
Prompt	Question given to the AI system.
AI Response	Raw AI-generated answer.
Generic Critique	Ordinary critique without HNS terminology.
HNS Error Type	Layer Jump, Scope Drift, Unsupported Causality, Metaphor Contamination, or Category Ambiguity.
Source Coordinate	Starting layer/category of the problematic claim.
Target Coordinate	Target layer/category or conclusion reached by the claim.
Diagnosis	Short structural explanation of the error.

Column	Description
Bridge Requirement	What logic would be required to repair the transition.
Severity	Low, medium, or high impact on reasoning reliability.
Evaluator	Human or AI evaluator identifier.

13. Conclusion

The HNS-36 PoC provides preliminary evidence that HNS can operate as a coordinate-based diagnostic language for AI-generated reasoning. The experiment shows that HNS can name structural defects, assign coordinates, and record recurring error patterns that ordinary critique often leaves unnamed or subjective.

The most important result is not that HNS produces better prose. The important result is that HNS can produce a more structured diagnostic record. This positions HNS as a backend verification layer for AI safety, red-team analysis, and external evaluation.

Final statement

HNS-36 should be advanced as a structural diagnostic layer: a system for naming, locating, and accumulating reasoning defects in AI outputs through a coordinate-based framework.

Appendix A. Compact HNS Evaluator Prompt

You are an HNS-36 structural evaluator. Your task is not to improve or rewrite the text. Your task is to diagnose structural reasoning errors in the text.

For each issue, identify:

1. Error Type: Layer Jump, Scope Drift, Unsupported Causality, Metaphor Contamination, or Category Ambiguity.
2. HNS Coordinate or transition path.
3. Source sentence or claim.
4. Structural diagnosis.
5. Required bridge logic.
6. Severity: low, medium, or high.

Return a table. Do not provide a revised answer unless separately requested.

Appendix B. Recommended 30-Case Domain Distribution

Domain	Number of Cases	Rationale
AI safety and hallucination	4	Direct relevance to external verification.
Education and inequality	3	Tests individual-to-societal explanation patterns.
Health and mental state explanation	4	Tests L1 to L3/L4 contamination risk.
Organizations and management	3	Tests L3/L5/L6 layer transition logic.
Climate and collective action	3	Tests societal-scale causal mechanisms.
Technology and social media	4	Tests design-intent to internal-state transitions.
Economics and public policy	3	Tests institutional and causal explanation patterns.
Human-AI interaction	3	Tests interface, cognition, and action categories.
Ethics and governance	3	Tests normative evaluation versus structural diagnosis.